

When carry crowds: endogenous foreign exchange crash risk, capital inflows, and the inefficiency of crowded carry

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Abstract

This paper develops a tractable model of crowded carry trades in which foreign capital inflows endogenously increase exchange-rate crash risk. A continuum of atomistic foreign investors chooses local-currency positions while taking aggregate inflows as given, although aggregate positioning raises both expected currency appreciation and the probability of a discrete depreciation. The symmetric competitive equilibrium is characterized by a cubic fixed-point equation with closed-form solutions. With positive baseline carry, the equilibrium is unique. When baseline carry is weak and the appreciation feedback is sufficiently strong, however, a no-inflow equilibrium can coexist with a self-sustaining inflow boom, so a deterioration in global financial conditions can trigger a discontinuous sudden stop. The competitive allocation is also generically inefficient because investors fail to internalize both the appreciation spillover and the systemic fragility created by their positions. In high-inflow states, fragility dominates and the market takes excessive tail risk. A Pigouvian macroprudential wedge reduces inflows and crash probability, implements the constrained-efficient allocation, and can eliminate fragile multiplicity. The framework provides a simple analytical link between the global financial cycle, crowded carry trades, endogenous tail risk, and macroprudential policy.

Keywords: Carry trades; Endogenous crash risk; Capital inflows; Sudden stops; Macroprudential policy

JEL Classification: F31, F32, F38, G15, G18

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1 Introduction

In the upswing of a global financial cycle, the trade that channels money into emerging markets (EMs) can look almost dull. A leveraged investor borrows cheaply in a major funding currency, buys a local-currency bond in an EM, and counts on the carry, the interest differential, plus a stable or appreciating exchange rate. For long stretches, nothing dramatic happens. Then, seemingly all at once, the same positions are cut, the exchange rate jumps, and what looked like a steady yield pick-up is revealed to be a strategy that earns small gains while occasionally losing big. This pattern, procyclical inflows in good global times and abrupt reversals when global conditions tighten, is by now a familiar description of portfolio flows and exchange-rate risk in EMs [Rey, 2015, Miranda-Agrippino and Rey, 2020, Forbes and Warnock, 2012, Fratzscher, 2012].

A central feature of this environment is that the “global financial cycle” is not just about the level of world interest rates. It is also about intermediaries’ risk appetite, balance-sheet capacity, and the price of bearing risk [Bruno and Shin, 2015, Adrian and Shin, 2010]. When global funding is easy and measured risk is low, cross-border balance sheets expand and gross capital flows surge; when funding tightens or risk appetite evaporates, the same balance sheets shrink and flows retrench [Broner et al., 2013, Forbes and Warnock, 2012]. Empirically, common “push” forces, global liquidity and global risk, explain a large share of the high-frequency variation in portfolio flows [Fratzscher, 2012, Ahmed and Zlate, 2014]. In that sense, the question for an EM is rarely whether it will be exposed to global shocks, but rather how that exposure maps into domestic asset prices, currency risk, and (crucially) the probability of disruptive tail events.

This paper offers a deliberately stripped-down framework for thinking about that mapping. The model keeps the basic carry-trade arithmetic but adds one element that is often discussed informally and modeled less often in closed form: tail risk is endogenous to positioning. When foreign portfolio inflows become large, they compress risk premia and support the currency in the short run, yet simultaneously increase fragility (through leverage, maturity mismatch, crowded positioning, or procyclical dealer balance sheets). As a result, the probability of a sharp exchange-rate crash rises with the scale of inflows. The same force that makes the trade look attractive in tranquil periods plants the seeds of a crash when conditions reverse.

A key conceptual choice is who controls the object that generates tail risk. We model a continuum of atomistic, identical foreign investors. Investor i chooses an individual position f_i and takes the aggregate inflow $F = \int f_i di$ as given. Crash risk and the equilibrium appreciation depend on F , not on any single f_i . This is the standard distinction between the individual control f and the aggregate state F familiar from

recursive competitive equilibrium (the “ (K, k) ” device): each agent is small relative to the market and therefore does not perceive that their own trade moves systemic risk, even though, in the symmetric equilibrium $f = F$, the aggregate of identical choices does exactly that. We show that with identical mean–variance preferences the continuum formulation and a representative-agent-with-aggregate formulation aggregate identically, so the two devices are interchangeable.

This decentralized reading changes the economics in three ways relative to a model in which a single agent internalizes F . First, because each investor treats crash risk and appreciation as given, the individual problem is linear in f and has a unique best response. The nonlinearity that makes carry crowding interesting reappears at the level of the equilibrium fixed point $f = F$, which is cubic in F . Tractability is therefore preserved: equilibrium inflows admit a closed-form (Cardano) characterization with transparent existence and uniqueness conditions. Second, the equilibrium is generically inefficient. A planner who internalizes that aggregate inflows raise crash risk and appreciate the currency would choose a different position. The wedge between the competitive and constrained-efficient allocations is closed-form and decomposes into a positive pecuniary appreciation externality and negative fragility externalities; its sign tells us whether the market over- or under-invests in tail risk. Third, macroprudential policy becomes genuinely Pigouvian: a wedge on foreigners’ after-cost return corrects the fragility externality rather than merely reshaping a single agent’s risk–return tradeoff.

The decentralized environment also clarifies the nature of multiplicity. When baseline carry is weak (so that inflows are unattractive at zero positioning) but the appreciation feedback is strong (so that large aggregate inflows make the carry attractive *ex post*), the aggregate best-response map crosses the 45° line more than once. The economy can sit at a no-inflow equilibrium or at a self-sustaining boom equilibrium. A small deterioration in global conditions, or a tightening of the policy wedge, can eliminate the boom equilibrium and trigger a discontinuous collapse: a sudden stop in the sense of [Calvo \[1998\]](#). Crucially, this multiplicity is a property of the aggregate fixed point and reflects a coordination feedback across atomistic agents; it is not the artifact of a single internalizing agent failing to select its global optimum. We are explicit that, with positive baseline carry, the competitive equilibrium is in fact unique; multiplicity is a knife-edge feature of the strong-feedback, weak-carry region.

The setup speaks to several strands of international finance. A first documents the global financial cycle and its consequences for monetary autonomy and cross-border leverage [[Rey, 2015](#), [Miranda-Agrippino and Rey, 2020](#), [Bruno and Shin, 2015](#)]. A second studies the macroeconomics of capital-flow bonanzas, stops, and crises [[Calvo, 1998](#), [Forbes and Warnock, 2012](#), [Gourinchas and Obstfeld, 2012](#), [Broner et al., 2013](#)].

A third focuses on currency carry trades and the empirical fact that carry returns are exposed to crash risk and negative skewness [Brunnermeier et al., 2009, Menkhoff et al., 2012, Jurek, 2014, Burnside et al., 2011, Lustig et al., 2011].

The closest conceptual connection is to the literature on inefficient credit cycles and pecuniary externalities. In Lorenzoni [2008], Bianchi [2011], and Mendoza [2010], atomistic borrowers do not internalize how their leverage affects equilibrium prices that enter collateral constraints, so competitive borrowing is excessive and a Pigouvian tax restores constrained efficiency [Jeanne and Korinek, 2010, Korinek, 2011, Bianchi and Mendoza, 2018, Farhi and Werning, 2016]. Our environment shares the externality logic but differs in the channel: alongside a pecuniary externality that operates through the exchange rate (as inflows appreciate the currency, in the spirit of the portfolio-balance and intermediary view of exchange rates in Gabaix and Maggiori, 2015), there is a non-pecuniary fragility externality by which aggregate inflows raise the probability of a discrete crash. The latter is closer to a systemic-risk externality and is, to our knowledge, novel in this closed-form form. Relative to disaster-risk approaches in which the disaster probability is an exogenous state [Barro, 2006, Gourio, 2012], here the crash probability is partly an equilibrium object: it is a function of the position the market chooses.

The paper makes three contributions. First, it provides a microfounded, decentralized model of carry inflows in which exchange-rate crash risk is endogenous to aggregate positioning, and shows that the competitive equilibrium is the fixed point $f = F$ of a linear individual problem, characterized by a cubic with closed-form roots and transparent uniqueness conditions. This addresses the central concern that internalization of F by a single agent is ad hoc: the nonlinearity now lives in the aggregate, not in any agent's perception.

Second, it characterizes the inefficiency of crowded carry. The wedge between the competitive and constrained-efficient allocations is closed-form, decomposes into appreciation and fragility externalities, and signs the direction of inefficiency. In the empirically relevant inflow-boom region, the market over-invests in tail risk, so a corrective wedge is welfare-improving. This connects the framework directly to the inefficient-credit-cycle literature and recasts macroprudential policy as Pigouvian.

Third, it delivers sharp policy predictions. A wedge on foreigners' after-cost return reduces equilibrium inflows and endogenous crash probability, can restore global concavity of the planner's objective and eliminate multiplicity, and admits a margin-style (Value-at-Risk) counterpart that binds precisely in the high-inflow region where jump risk steepens the variance. We also record clean comparative statics with respect to the global financial cycle under monotonicities that we motivate with reference to the empirical and theoretical literature.

Section 2 presents the decentralized environment and the exchange-rate process with aggregate-dependent crash probability, derives the individual problem and the equilibrium condition, establishes existence, uniqueness, and closed-form characterization, and introduces the constrained-efficient benchmark and the externality wedge. It then studies multiplicity and self-fulfilling sudden stops, the risk-premium and tail-risk objects, macroprudential policy, the connection to inefficient credit cycles, and global-cycle comparative statics. Section 3 concludes. All proofs are collected in Appendix A; Appendix B develops the margin-style instrument.

2 Model

2.1 Set up

Time is $t \in \{0, 1\}$. There is a unit continuum of identical, atomistic foreign investors indexed by $i \in [0, 1]$, and a single emerging-market (EM) local-currency security. Throughout, variables are expressed in gross form.

Let $e_t > 0$ denote the spot exchange rate, defined as foreign currency per one unit of EM currency, so that an increase in e is an EM appreciation from the foreign investor's point of view. A scalar global state $g \in (0, 1]$ summarizes the global financial cycle; higher g means easier global conditions (lower funding frictions and higher effective risk tolerance). The global state affects: i) the foreign funding gross rate $R^*(g) > 1$; ii) the investors' mean-variance risk-aversion coefficient $\lambda(g) > 0$; iii) the baseline volatility of normal-times currency fluctuations $\sigma_\varepsilon(g) > 0$; and iv) the baseline crash probability $\bar{\pi}(g) \in (0, 1)$. Sign restrictions on these functions are imposed in Section 2.11 and motivated there.

At $t = 0$ investor i allocates an amount $f_i \geq 0$ (in foreign-currency units) into the local-currency security, which has gross local-currency return $R > 1$ between $t = 0$ and $t = 1$. The aggregate inflow is

$$F \equiv \int_0^1 f_i di.$$

The object R can be read as the gross return on a one-period EM local-currency bond or a broad EM local-currency index.¹

¹Keeping R as a primitive matches the logic of the model: the core object is the foreign-currency excess payoff that combines the local return and the exchange-rate change. One may endogenize R or the bond price through market clearing; we discuss this extension in the concluding remarks and note that a price channel would reinforce the amplification mechanism.

Investor i funds f_i at the global gross rate $R^*(g)$. A host-country macroprudential wedge $\tau \in [0, 1)$ acts like a tax on the local return component received by foreigners.² The investor's foreign-currency net excess payoff (relative to funding) is

$$K_i \equiv f_i \left[(1 - \tau) \frac{e_1}{e_0} R - R^*(g) \right] - \frac{\kappa}{2} f_i^2, \quad (1)$$

where e_1/e_0 is the gross currency appreciation factor and $\kappa > 0$ is a convex balance-sheet or price-impact cost (limited depth, intermediation cost, or internal leverage penalties).

2.2 The exchange rate and the aggregate fragility channel

The exchange-rate change contains a normal innovation and a crash jump whose probability depends on aggregate inflows:

$$\frac{e_1}{e_0} = \underbrace{1 + \mu_e(g) + \chi F + \varepsilon}_{\text{normal component}} - \underbrace{J \mathbf{1}\{\mathcal{C} = 1\}}_{\text{crash jump}}, \quad (2)$$

where $\mu_e(g) \in \mathbb{R}$ is baseline expected appreciation; $\chi \geq 0$ is a portfolio-balance channel by which larger aggregate inflows tilt demand toward EM currency and raise expected appreciation (imperfect sterilization, slow-moving arbitrage, intermediary risk-bearing as in [Gabaix and Maggiori, 2015](#)); ε is a normal-times shock with $\mathbb{E}[\varepsilon] = 0$ and $\text{Var}(\varepsilon) = \sigma_\varepsilon^2(g)$; and $\mathcal{C} \in \{0, 1\}$ is a crash indicator. If $\mathcal{C} = 1$, the EM currency suffers a discrete depreciation of gross size $J > 0$.³

The crash probability is endogenous to the aggregate position:

$$\mathbb{P}(\mathcal{C} = 1 | F, g) = \pi(F, g) \equiv \bar{\pi}(g) + \eta F^2, \quad (3)$$

with $\eta > 0$ measuring fragility: larger aggregate inflows raise the likelihood of an FX crash (currency mismatch, fragile funding, crowded carry, procyclical dealer balance sheets). The quadratic form is the simplest smooth specification that generates strong nonlinearities while keeping the equilibrium polynomial.

Remark 1 (The individual–aggregate distinction). Both the appreciation term χF in Equation (2) and the crash probability $\pi(F, g)$ in Equation (3) depend on the aggregate F , not on the individual f_i . Each investor is of measure zero and therefore perceives π

²This wedge is deliberately reduced-form: it can capture a tax on foreign holdings, an unremunerated reserve requirement, or a regulatory wedge that lowers the effective after-cost return to foreign intermediaries.

³Because e is foreign per EM, a crash means e_1/e_0 falls; hence the subtraction in Equation (2).

and e_1/e_0 as exogenous when choosing f_i . In the symmetric equilibrium $f_i = F$ for all i , however, the common choice and the aggregate coincide. This is the precise sense in which carry crowding is an aggregate phenomenon that no individual controls.

Assumption 1 (Feasible crash probability). There exists $\bar{F} > 0$ such that the equilibrium inflow satisfies $F \in [0, \bar{F}]$ and $\pi(F, g) \in (0, 1)$ for all (F, g) in this set.

Remark 2. Assumption 1 is mild. Since $\pi(F, g) = \bar{\pi}(g) + \eta F^2$, one can pick $\bar{F}$ such that $\eta \bar{F}^2 < 1 - \sup_g \bar{\pi}(g)$.

2.3 The individual problem and the best response

Investor i is mean–variance. Let $\theta > 0$ be a taste shifter on expected payoff (a “mean weight”) and $\lambda(g) > 0$ risk aversion. Investor i chooses $f_i \geq 0$ to maximize

$$U_i(f_i; F, g, \tau) \equiv \theta \mathbb{E}[K_i | f_i, F, g] - \frac{\lambda(g)}{2} \text{Var}(K_i | f_i, F, g), \quad (4)$$

taking the aggregate F as given. Higher θ increases desired exposure; higher $\lambda(g)$ reduces it, and λ moves with global conditions.

Define the random per-unit excess return in foreign currency, common to all investors and a function of the aggregate state,

$$X(F, g, \tau) \equiv (1 - \tau) \frac{e_1}{e_0} R - R^*(g). \quad (5)$$

Then $K_i = f_i X - \frac{\kappa}{2} f_i^2$, and conditional on (F, g, τ) ,

$$\mathbb{E}[K_i] = f_i \mu_X(F, g, \tau) - \frac{\kappa}{2} f_i^2, \quad \text{Var}(K_i) = f_i^2 \sigma_X^2(F, g, \tau), \quad (6)$$

where $\mu_X = \mathbb{E}[X]$ and $\sigma_X^2 = \text{Var}(X)$ depend only on the aggregate F . Using Equation (2)–(3) and the independence of ε and $\mathcal{C}$,

$$\mu_X(F, g, \tau) = (1 - \tau) R (1 + \mu_e(g) + \chi F - J \pi(F, g)) - R^*(g), \quad (7)$$

$$\sigma_X^2(F, g, \tau) = (1 - \tau)^2 R^2 \left(\sigma_\varepsilon^2(g) + J^2 \pi(F, g) (1 - \pi(F, g)) \right). \quad (8)$$

The term $J \pi(F, g)$ in Equation (7) is a crash-expected-loss adjustment; $J^2 \pi(1 - \pi)$ in Equation (8) is the jump contribution to variance. For closed-form characterization we use the standard rare-crash approximation.

Assumption 2 (Rare crash). In the relevant equilibrium region, $\pi(F, g)$ is small enough

that $1 - \pi(F, g) \approx 1$ in the second-order term of Equation (8), so that

$$\sigma_X^2(F, g, \tau) \approx (1 - \tau)^2 R^2 (\sigma_\varepsilon^2(g) + J^2 \pi(F, g)). \quad (9)$$

Remark 3. Assumption 2 preserves the qualitative economics while yielding a polynomial equilibrium condition. Dropping it and keeping $\pi(1 - \pi)$ leaves the proofs intact at the cost of heavier algebra (the equilibrium polynomial rises in degree).

It is convenient to collect the following composite coefficients, exactly as in the reduced-form, now understood as functions of the aggregate state:

$$A(g, \tau) \equiv (1 - \tau)R(1 + \mu_e(g) - J\bar{\pi}(g)) - R^*(g), \quad (10)$$

$$B(\tau) \equiv (1 - \tau)R\chi, \quad (11)$$

$$C(g, \tau) \equiv (1 - \tau)R J\eta, \quad (12)$$

$$S(g, \tau) \equiv (1 - \tau)^2 R^2 (\sigma_\varepsilon^2(g) + J^2 \bar{\pi}(g)), \quad (13)$$

$$D(\tau) \equiv (1 - \tau)^2 R^2 J^2 \eta. \quad (14)$$

Here A is baseline expected excess carry (net of baseline crash losses); B is the strength of inflow-driven expected appreciation; C is the marginal expected crash loss induced by inflows; S is baseline variance; and D is how strongly inflows raise variance via endogenous crash probability. With these, the conditional mean and variance of X are

$$\mu_X(F, g, \tau) = A + BF - CF^2, \quad \sigma_X^2(F, g, \tau) = S + DF^2. \quad (15)$$

Investor i 's objective becomes

$$U_i(f_i; F, g, \tau) = \theta \left(f_i \mu_X(F, g, \tau) - \frac{\kappa}{2} f_i^2 \right) - \frac{\lambda(g)}{2} f_i^2 \sigma_X^2(F, g, \tau). \quad (16)$$

Because μ_X and σ_X^2 are taken as given, U_i is strictly concave quadratic in f_i , with a unique best response.

Lemma 1 (Linear best response). *For any (F, g, τ) , investor i 's problem (16) is strictly concave in f_i , and the unique best response is*

$$\mathcal{B}(F; g, \tau) = \max \left\{ 0, \frac{\theta \mu_X(F, g, \tau)}{\theta \kappa + \lambda(g) \sigma_X^2(F, g, \tau)} \right\}. \quad (17)$$

Proof. See Appendix A. □

The economics is transparent. The atomistic investor equates the marginal expected gain $\theta \mu_X$ to the marginal cost from the balance-sheet penalty $\theta \kappa f_i$ plus the marginal

risk penalty $\lambda\sigma_X^2 f_i$. Crash risk and appreciation enter only through μ_X and σ_X^2 , which the investor cannot move. This is precisely where the model departs from one in which a single agent internalizes F : the convex feedback of inflows on tail risk is absent from the individual first-order condition and reappears only when we close the model in the aggregate.

2.4 Competitive equilibrium: the cubic that governs everything

We use the symmetric competitive equilibrium throughout. It is the natural decentralization of the carry trade: identical small investors best respond to an aggregate they do not control, and consistency requires that the common choice equal the aggregate.

Definition 1 (Symmetric competitive equilibrium). A symmetric competitive equilibrium (CE) is an inflow level $F^{\text{CE}}(g, \tau) \geq 0$ such that every investor chooses $f_i = F^{\text{CE}}$ optimally given the aggregate $F = F^{\text{CE}}$; that is, $F^{\text{CE}} = \mathcal{B}(F^{\text{CE}}; g, \tau)$.

Imposing $f_i = F$ on the best response (17) (interior case $\mu_X > 0$) gives $F(\theta\kappa + \lambda\sigma_X^2(F)) = \theta\mu_X(F)$. Substituting (15) and rearranging yields the equilibrium cubic:

$$\lambda(g)D(\tau)F^3 + \theta C(g, \tau)F^2 + [\theta\kappa + \lambda(g)S(g, \tau) - \theta B(\tau)]F - \theta A(g, \tau) = 0. \quad (18)$$

We write Equation (18) compactly as $H(F; g, \tau) = 0$ with

$$H(F; g, \tau) \equiv a_3 F^3 + a_2 F^2 + a_1 F + a_0, \quad \begin{aligned} a_3 &= \lambda(g)D(\tau) > 0, & a_2 &= \theta C(g, \tau) > 0, \\ a_1 &= \theta\kappa + \lambda(g)S(g, \tau) - \theta B(\tau), & a_0 &= -\theta A(g, \tau). \end{aligned} \quad (19)$$

It is worth pausing on the contrast with a single internalizing agent. If one agent chose F knowing that π and e_1/e_0 respond to F , the stationarity condition would carry the extra terms produced by differentiating $\mu_X(F)F$ and $\sigma_X^2(F)F^2$ through F . Those extra terms are exactly the externality the atomistic investor ignores, and we make them explicit in Section 2.6. The following subsection records existence, uniqueness, and the closed form for the competitive cubic (18); the economic interpretation of each result is stated in the main text, with proofs deferred to Appendix A.

2.5 Existence, uniqueness, and closed form

Lemma 2 (Existence of an interior competitive equilibrium). *Suppose $a_3 > 0$ (i.e. $\lambda(g), D(\tau) > 0$) and $A(g, \tau) > 0$. Then $H(0; g, \tau) < 0$ and $H(F; g, \tau) \rightarrow +\infty$ as $F \rightarrow \infty$, so (18)*

has at least one strictly positive root, and there exists an interior competitive equilibrium $F^{\text{CE}}(g, \tau) \in (0, \infty)$.

Proof. See Appendix A. □

The condition $A > 0$ says that baseline carry is attractive even at zero positioning, so the market does not collapse to no inflows. The economics of the limit is the mirror image of the single-agent intuition: as aggregate inflows grow without bound, equilibrium variance scales with F^2 (through DF^2) and crash-loss drag scales with F^2 (through CF^2), so the marginal investor's perceived return turns negative and inflows cannot be sustained.

Proposition 1 (Uniqueness with positive baseline carry). *If $A(g, \tau) > 0$, the competitive equilibrium is unique. More generally, a sufficient condition for at most one positive root of (18) is $a_2 > 0$ and $a_0 \leq 0$, which holds whenever $C > 0$ and $A \geq 0$.*

Proof. See Appendix A. □

The proof (Appendix A) is a one-line application of Descartes' rule of signs to (19): with $a_3 > 0$, $a_2 > 0$, $a_0 < 0$, and a_1 of either sign, the coefficient sequence exhibits exactly one sign change, so there is exactly one positive root. This delivers a clean and economically important message that the convexity of crash risk in inflows steepens the cost of large positions but does not, on its own, create a second equilibrium. Multiplicity requires a different configuration, analyzed in Section 2.7.

Closed-form roots. Write Equation (18) in the form $a_3 F^3 + a_2 F^2 + a_1 F + a_0 = 0$ and apply the depressed-cubic (Tschirnhaus) transformation $F = y - a_2/(3a_3)$ to obtain

$$y^3 + p y + q = 0, \quad p = \frac{3a_3 a_1 - a_2^2}{3a_3^2}, \quad q = \frac{2a_2^3 - 9a_3 a_2 a_1 + 27a_3^2 a_0}{27a_3^3}, \quad (20)$$

with discriminant $\Delta = (q/2)^2 + (p/3)^3$.

Proposition 2 (Characterization of the real roots). *The depressed cubic (20) has exactly one real root if $\Delta > 0$, three distinct real roots if $\Delta < 0$, and a repeated real root if $\Delta = 0$. Any real root y maps to a real F via $F = y - a_2/(3a_3)$. When $\Delta > 0$ the unique real root is*

$$y^* = \sqrt[3]{-\frac{q}{2} + \sqrt{\Delta}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\Delta}}, \quad F^{\text{CE}} = y^* - \frac{a_2}{3a_3}, \quad (21)$$

and when $\Delta < 0$ the three real roots admit the trigonometric representation

$$y_k = 2\sqrt{-\frac{p}{3}} \cos\left(\frac{1}{3} \arccos\left(\frac{3q}{2p}\sqrt{-\frac{3}{p}}\right) - \frac{2\pi k}{3}\right), \quad k = 0, 1, 2, \quad (22)$$

with $F_k = y_k - a_2/(3a_3)$. The economically relevant equilibria are the nonnegative roots; in the multiplicity case (Section 2.7) stability selects among them.

Proof. See Appendix A. □

2.6 Constrained efficiency and the inefficiency of crowded carry

We now make precise the sense in which the competitive equilibrium is inefficient, and we recover the earlier version's quartic objective as the planner's problem. Consider a constrained planner who chooses the common position $f = F$ while internalizing that aggregate inflows move both the crash probability and the appreciation. We treat the macroprudential wedge as a pure transfer that is rebated lump-sum, so welfare is evaluated using the representative investor's own mean-variance criterion with the externality internalized.⁴

The planner's objective is

$$W(F; g, \tau) = \theta \left(F \mu_X(F, g, \tau) - \frac{\kappa}{2} F^2 \right) - \frac{\lambda(g)}{2} F^2 \sigma_X^2(F, g, \tau), \quad (23)$$

which, substituting (15), is the quartic

$$W(F; g, \tau) = \theta \left(AF + BF^2 - CF^3 - \frac{\kappa}{2} F^2 \right) - \frac{\lambda(g)}{2} F^2 (S + DF^2). \quad (24)$$

Its first-order condition is the social cubic

$$W_F(F; g, \tau) = \theta (A + (2B - \kappa)F - 3CF^2) - \lambda(g)(SF + 2DF^3) = 0. \quad (25)$$

Equation (25) is precisely the stationarity condition that an agent internalizing F would face. The difference between Equation (25) and the private market-clearing condition (18) is the externality.

Proposition 3 (The externality wedge). *Let the marginal private value of inflows, evaluated*

⁴This is the standard constrained-efficiency benchmark in the pecuniary-externality literature [Lorenzoni, 2008, Bianchi, 2011]: the planner respects the same preferences and market structure as private agents but internalizes the dependence of equilibrium objects on the aggregate. The benchmark inherits the mean-variance functional rather than a deep utility primitive; we view this as the natural normative counterpart of the positive model and discuss a fuller welfare treatment in Section 3.

in the symmetric configuration $f_i = F$, be

$$\text{MV}^{\text{priv}}(F) \equiv \theta \mu_X(F) - \theta \kappa F - \lambda(g) F \sigma_X^2(F),$$

so that $\text{MV}^{\text{priv}}(F) = 0$ is the competitive condition (18). Then the marginal social value W_F exceeds the marginal private value by

$$\delta(F) \equiv W_F(F) - \text{MV}^{\text{priv}}(F) = \theta B F - 2\theta C F^2 - \lambda(g) D F^3 = F(\theta B - 2\theta C F - \lambda(g) D F^2). \quad (26)$$

Proof. See Appendix A. □

The wedge $\delta(F)$ decomposes the inefficiency into three economically distinct pieces: 1) $+\theta B F$: a *positive pecuniary externality*. Each unit of inflow appreciates the currency (through χ) and thereby raises the carry earned by all other investors. Atomistic investors ignore this, which pushes the competitive allocation below the efficient one. This is the appreciation analogue of the relative-price externalities in Lorenzoni [2008] and Gabaix and Maggiori [2015]; 2) $-2\theta C F^2$: a *negative fragility externality in the mean*. Each unit of inflow raises the systemic crash probability and hence the expected crash loss borne by everyone. Ignored privately, this pushes the competitive allocation above the efficient one; 3) $-\lambda(g) D F^3$: a *negative fragility externality in the variance*. Inflows raise systemic jump risk and therefore the risk penalty everyone pays. Again ignored privately, pushing toward over-investment.

Proposition 4 (Direction of inefficiency). *At the competitive equilibrium F^{CE} , $W_F(F^{\text{CE}}) = \delta(F^{\text{CE}})$. Define the threshold*

$$\hat{F} \equiv \frac{-\theta C + \sqrt{\theta^2 C^2 + \lambda(g) D \theta B}}{\lambda(g) D} > 0, \quad (27)$$

the unique positive root of $\lambda(g) D F^2 + 2\theta C F - \theta B = 0$. Then:

1. *if $F^{\text{CE}} > \hat{F}$, the fragility externalities dominate, $\delta(F^{\text{CE}}) < 0$, and the competitive equilibrium over-invests relative to the constrained-efficient allocation F^{SP} (so $F^{\text{SP}} < F^{\text{CE}}$);*
2. *if $F^{\text{CE}} < \hat{F}$, the appreciation externality dominates, $\delta(F^{\text{CE}}) > 0$, and the competitive equilibrium under-invests ($F^{\text{SP}} > F^{\text{CE}}$).*

Proof. See Appendix A. □

The substantive content is that crowding is inefficient, and the direction of inefficiency is state-dependent. In tranquil periods with modest positioning ($F^{\text{CE}} < \hat{F}$), the dominant uninternalized effect is the beneficial appreciation spillover, and competition

leaves inflows inefficiently low. In inflow booms ($F^{\text{CE}} > \hat{F}$), precisely the empirically relevant region in which gross flows surge with easy global conditions, the dominant uninternalized effect is the fragility spillover, and competition produces inefficiently large positions and excessive tail risk. This is the carry-trade counterpart of the over-borrowing result in Bianchi [2011] and Mendoza [2010], and it is what gives macro-prudential policy a Pigouvian rationale.

Proposition 5 (A Pigouvian wedge implements the efficient allocation). *Suppose at $\tau = 0$ the competitive equilibrium over-invests, $F^{\text{CE}}(g, 0) > F^{\text{SP}}(g)$, where $F^{\text{SP}}(g) = \arg \max_F W(F; g, 0)$ is the constrained-efficient allocation evaluated at gross (pre-transfer) returns. Since $\partial F^{\text{CE}} / \partial \tau < 0$ (Proposition 7) and F^{CE} is continuous in τ , there exists a wedge $\tau^* \in (0, 1)$ such that $F^{\text{CE}}(g, \tau^*) = F^{\text{SP}}(g)$. The wedge τ^* corrects the fragility externality: it lowers the private marginal gain from building positions while leaving the externality channel intact, so the decentralized allocation coincides with the constrained-efficient one.*

Proof. See Appendix A. □

Three points deserve emphasis. First, the macroprudential wedge here is not merely a device that reshapes a single agent's risk–return tradeoff; it corrects a genuine externality that atomistic investors cannot be expected to internalize. Second, because the wedge operates on the after-cost return, it is robust: it does not require the regulator to identify every micro source of fragility, only to lean against the scale of crowded positioning. Third, the very same instrument can change the number of equilibria, to which we now turn.

2.7 Multiple equilibria and self-fulfilling sudden stops

Proposition 1 shows that with positive baseline carry the competitive equilibrium is unique. Multiplicity is therefore not a generic feature of crowded carry; it arises in a specific and economically interpretable region.

Theorem 1 (Self-fulfilling sudden stops). *Suppose baseline carry is unattractive at zero positioning, $A(g, \tau) < 0$, and the appreciation feedback is strong relative to balance-sheet and baseline-risk forces,*

$$\theta B(\tau) > \theta \kappa + \lambda(g) S(g, \tau), \quad (28)$$

i.e. $a_1 < 0$ in (19). Then $F = 0$ is a competitive equilibrium, and there exist parameter values for which Equation (18) additionally has two positive roots $0 < F_L < F_H$. The low positive root F_L is unstable under best-response dynamics, while $F = 0$ and F_H are stable. Consequently the economy admits a no-inflow equilibrium and a self-sustaining high-inflow (boom) equilibrium.

A small deterioration in global conditions (lower g) or a tightening of the wedge (higher τ) can eliminate F_H , producing a discontinuous collapse of inflows—a sudden stop.

Proof. See Appendix A. □

The mechanism is a coordination feedback operating through the aggregate, not the indecision of a single internalizing agent. When $A < 0$, no investor wants to enter if no one else does, so $F = 0$ is self-confirming. When condition (28) holds, however, a large enough aggregate position appreciates the currency by enough to make the carry attractive ex post, so $F_H > 0$ is also self-confirming. The unstable interior root F_L is the tipping point between the two. This is the carry-trade version of the multiple-equilibria sudden stops of Calvo [1998], and it places the multiplicity squarely in the aggregate, addressing the concern that a single representative internalizer would simply pick its global maximum: there is no single optimizer here, only a market that can settle on more than one mutually consistent outcome.

Corollary 1 (Macprudential wedges can remove multiplicity). *Fix g . If τ is raised enough that condition (28) fails, i.e. $\theta B(\tau) \leq \theta\kappa + \lambda(g)S(g, \tau)$, then $a_1 \geq 0$ and the only equilibria are: the unique interior $F^{\text{CE}} > 0$ if $A > 0$, or $F = 0$ if $A \leq 0$. In particular the boom equilibrium F_H can no longer coexist with the no-inflow equilibrium, so the discontinuous switching of Theorem 1 is ruled out. The wedge stabilizes the regime structure by dampening the appreciation feedback $B(\tau) = (1 - \tau)R\chi$.*

Proof. See Appendix A. □

The policy reading is that the regulator’s task is not only to lower the average level of inflows but to keep the economy out of the configuration in which the equilibrium set is fragile. Because $B(\tau)$ falls in τ , a wedge that is large enough during strong-feedback episodes removes the boom equilibrium’s dependence on continued coordination and thereby removes a source of discontinuous reversals.

2.8 Risk price, tail risk, and amplification

A risk-price analogue. Summarize incentives at the aggregate position by a risk price (mean per unit of risk),

$$\rho(F, g, \tau) \equiv \frac{\mu_X(F, g, \tau)}{\sigma_X(F, g, \tau)} = \frac{A + BF - CF^2}{\sqrt{S + DF^2}}. \quad (29)$$

The numerator is carry net of endogenous crash losses; the denominator is total risk (normal volatility plus jump risk that rises with inflows).

Lemma 3 (Endogenous tail risk lowers the risk price at high inflows). *If $C > 0$ and $D > 0$, then $\rho(F, g, \tau) \rightarrow -\infty$ as $F \rightarrow \infty$, and there exists $\bar{F}_\rho > 0$ such that ρ is strictly decreasing for all $F \geq \bar{F}_\rho$.*

Proof. See Appendix A. □

The expected return declines quadratically in F (through ηF^2) while volatility grows only linearly, so the crash drag eventually dominates and the per-unit reward to carry collapses in crowded markets.

Tail-risk elasticity. Define the tail event as the crash state and the tail probability $\mathcal{T}(F, g) \equiv \pi(F, g) = \bar{\pi}(g) + \eta F^2$.

Proposition 6 (Inflow elasticity of crash probability). *The log–log elasticity of tail risk with respect to aggregate inflows is*

$$\varepsilon_{\mathcal{T},F}(F, g) \equiv \frac{\partial \log \mathcal{T}}{\partial \log F} = \frac{2\eta F^2}{\bar{\pi}(g) + \eta F^2}. \quad (30)$$

If $\eta F^2 \gg \bar{\pi}(g)$, then $\varepsilon_{\mathcal{T},F} \rightarrow 2$: a 1% increase in aggregate inflows raises crash probability by about 2%.

Proof. See Appendix A. □

This convex sensitivity is the heart of the policy concern: once the economy is far enough into the high-inflow region, incremental inflows mainly buy tail risk even while average returns still look attractive.

Amplification, internalized vs. not. The competitive cubic (18) embeds two opposing aggregate feedbacks: carry amplification (inflows raise appreciation, term θBF) and fragility (inflows raise crash loss and variance, terms θCF^2 and λDF^3). The social cubic (25) embeds the same feedbacks but with the coefficients $2\theta B$, $3\theta C$, and $2\lambda D$, reflecting full internalization. The contrast is instructive: every place the social condition carries a larger coefficient is a place where the private market under-weights the marginal effect of its own crowding. The wedge $\delta(F)$ of Proposition 3 is exactly the sum of these coefficient gaps multiplied by the relevant powers of F .

2.9 Connection to inefficient credit cycles

The model is, in its normative content, a member of the family of economies with constrained-inefficient borrowing booms. In [Lorenzoni \[2008\]](#), atomistic entrepreneurs do not internalize that their leverage depresses asset prices in bad states, tightening collateral constraints for all; in [Bianchi \[2011\]](#) and [Mendoza \[2010\]](#), individual borrowing does not internalize its effect on the equilibrium relative price that enters a flow collateral constraint; in each case competitive borrowing is excessive and a state-contingent tax restores efficiency [[Jeanne and Korinek, 2010](#), [Korinek, 2011](#), [Bianchi and Mendoza, 2018](#)].

Our economy maps onto this structure with two transmission channels rather than one. The appreciation channel χ is a *pecuniary* externality of the standard type: it operates through the equilibrium exchange rate, which is itself moved by aggregate flows under limited risk-bearing capacity [[Gabaix and Maggiori, 2015](#)]. The fragility channel η is a *non-pecuniary, systemic-risk* externality: aggregate positioning raises the probability of a discrete crash that hits all investors, in the spirit of the aggregate-demand and risk externalities emphasized by [Farhi and Werning \[2016\]](#) and [Korinek \[2011\]](#). The wedge (26) shows how the two combine. The contribution relative to this literature is to deliver the externality, its sign, and the corrective instrument in closed form within a transparent carry-trade environment, and to show that the same instrument that corrects the externality can also collapse a fragile multiplicity. We view this as a useful baseline on which richer general-equilibrium treatments can build.

2.10 Comparative statics: the macroprudential wedge

We now record sharp derivatives at an interior competitive equilibrium. The tool is the implicit function theorem applied to $H(F; g, \tau) = 0$ in (19); at the unique interior CE with $A > 0$ we have $H_F(F^{\text{CE}}) > 0$.

Proposition 7 (Higher τ lowers inflows and tail risk). *At an interior competitive equilibrium with $H_F(F^{\text{CE}}; g, \tau) > 0$,*

$$\frac{\partial F^{\text{CE}}}{\partial \tau} < 0, \quad \frac{\partial \mathcal{T}(F^{\text{CE}}, g)}{\partial \tau} < 0. \quad (31)$$

Proof. See Appendix A. □

The proof (Appendix A) shows that the equilibrium condition can be written so that the relevant cross-partial reduces, after substituting the equilibrium risk premium, to a term proportional to $(\text{Risk Premium} - R^* - \kappa F)$, which is negative because the gross

funding cost exceeds an equilibrium risk premium of the order of a few percent. Intuitively, the wedge lowers the after-cost carry one-for-one while affecting the risk penalty only at second order, so the marginal investor scales back, the aggregate falls, and, because tail risk is increasing in the aggregate, the crash probability falls with it.

2.11 Comparative statics: the global financial cycle

To connect to the global financial cycle we impose standard monotonicities, which we now motivate explicitly in response to the natural question of why these particular functions of g move as assumed.

Assumption 3 (Global-cycle monotonicities). For higher g (easier global conditions),

$$\frac{dR^*(g)}{dg} < 0, \quad \frac{d\lambda(g)}{dg} < 0, \quad \frac{d\bar{\pi}(g)}{dg} < 0, \quad \frac{d\sigma_\varepsilon(g)}{dg} \leq 0, \quad \frac{d\mu_e(g)}{dg} \geq 0.$$

These restrictions are the reduced-form footprint of well-documented mechanisms. The decline of the funding rate $R^*(g)$ in easy conditions is the defining feature of the global liquidity and U.S. monetary transmission documented by [Bruno and Shin \[2015\]](#) and [Miranda-Agrippino and Rey \[2020\]](#). The decline of effective risk aversion $\lambda(g)$ is the risk-taking channel of [Adrian and Shin \[2010\]](#) and [Bruno and Shin \[2015\]](#) and the intermediary-balance-sheet view of [He and Krishnamurthy \[2013\]](#); empirically it tracks the global risk-appetite factor proxied by the VIX in [Rey \[2015\]](#) and [Miranda-Agrippino and Rey \[2020\]](#). The decline of baseline crash probability $\bar{\pi}(g)$ and of normal-times volatility $\sigma_\varepsilon(g)$ in tranquil global states is consistent with the surge/retranchment dynamics of [Forbes and Warnock \[2012\]](#), the comovement of FX volatility with the global cycle in [Menkhoff et al. \[2012\]](#), and the disaster-probability logic of [Gourinchas and Obstfeld \[2012\]](#). Finally, the (weak) rise of baseline appreciation $\mu_e(g)$ reflects the financial channel of the exchange rate by which easy global conditions and inflows support EM currencies [[Gabaix and Maggiori, 2015](#)].

Proposition 8 (Easier global conditions raise inflows). *Under Assumption 3, at an interior competitive equilibrium with $H_F(F^{\text{CE}}; g, \tau) > 0$,*

$$\frac{\partial F^{\text{CE}}}{\partial g} > 0 \quad \text{in parameter regions where baseline carry } A(g, \tau) \text{ rises in } g. \quad (32)$$

Proof. See Appendix [A](#). □

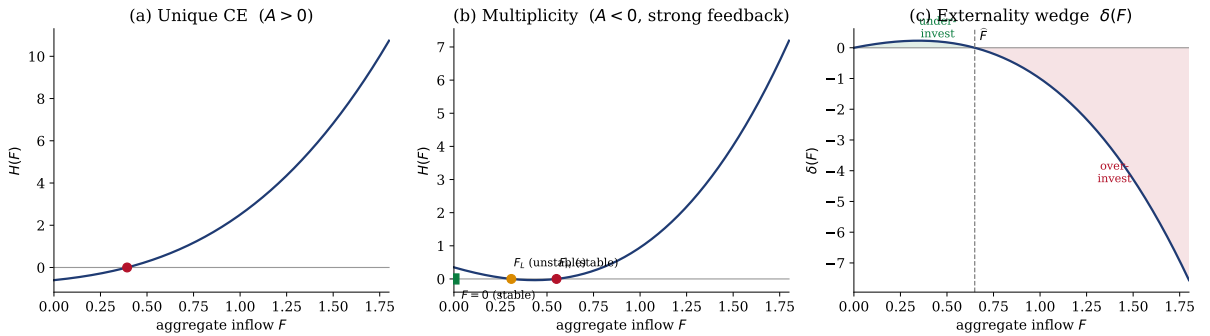
The channels are the natural ones: a higher g lowers the funding cost and baseline crash drag (raising A), lowers risk aversion, and lowers baseline variance, all of which

raise the marginal investor's desired position and hence the aggregate. Combined with Proposition 6, easier global conditions raise inflows and—through the convex tail-risk elasticity—raise crash probability more than proportionately, which is why a countercyclical wedge that leans against g is valuable.

2.12 A numerical illustration

Figure 1 illustrates the three central objects on a common, deliberately stylized parameterization, plotting the equilibrium polynomial $H(F)$ from (19) and the wedge $\delta(F)$ from (26). Panel (a) takes positive baseline carry ($a_0 = -\theta A < 0$): H is negative at the origin and crosses zero exactly once, so the competitive equilibrium is unique, consistent with Proposition 1. Panel (b) takes weak baseline carry ($A < 0$, so $a_0 > 0$) together with the strong-feedback condition (28) ($a_1 < 0$): now $F = 0$ is an equilibrium, H dips below zero, and there are two additional positive roots—an unstable tipping point F_L and a stable boom F_H , exactly the configuration of Theorem 1. A small adverse move in g or a tightening of τ lifts the local minimum of H above zero, the pair (F_L, F_H) disappears, and inflows collapse discontinuously to $F = 0$. Panel (c) plots the externality wedge: $\delta(F) > 0$ for $F < \hat{F}$ (the appreciation externality dominates and the market under-invests) and $\delta(F) < 0$ for $F > \hat{F}$ (the fragility externalities dominate and the market over-invests), as in Proposition 4. The vertical line marks the threshold $\hat{F}$ of (27).

Figure 1: Equilibrium and inefficiency in the decentralized carry economy



Panel (a): with positive baseline carry the equilibrium polynomial $H(F)$ crosses zero once (unique competitive equilibrium). Panel (b): with weak baseline carry and strong appreciation feedback, $F = 0$, an unstable interior root F_L , and a stable boom F_H coexist; a small deterioration in conditions eliminates F_H (a self-fulfilling sudden stop). Panel (c): the externality wedge $\delta(F)$ is positive (under-investment) below the threshold $\hat{F}$ and negative (over-investment) above it. Parameters are illustrative and chosen only to display the geometry.

The figure makes the policy logic concrete. The economically dangerous region is the right portion of panel (c), where each additional unit of crowded carry mainly buys systemic tail risk that the marginal investor does not price. A wedge τ both slides

the competitive root in panel (a) leftward (Proposition 7) and, in the configuration of panel (b), removes the boom equilibrium by flattening the appreciation feedback (Corollary 1). The same instrument therefore addresses the level inefficiency and the regime fragility at once.

2.13 Beyond exchange rates

Remark 4 (General crowded trades). Nothing in the analysis is specific to currencies. Reinterpret e_1/e_0 as the gross return on any risky asset whose crash probability rises with aggregate positioning—crowded credit, equity-momentum, or volatility-selling strategies—and let χ capture any price-pressure feedback by which aggregate buying supports the asset in normal times. Equations (18)–(26) are unchanged, as are the existence, uniqueness, externality, and policy results. The framework is thus a general baseline for the inefficiency of crowded trades, with the FX application as a leading case.

3 Conclusion

This paper studies a decentralized carry-trade economy in which the probability of an FX crash is endogenous to the aggregate scale of portfolio inflows, $\pi(F, g) = \bar{\pi}(g) + \eta F^2$. A continuum of atomistic investors chooses individual positions f taking the aggregate F as given, and the symmetric competitive equilibrium $f = F$ is the fixed point of a linear individual problem. Two features make the environment tractable yet rich. The individual problem is concave with a unique best response, so the nonlinearity of carry crowding lives entirely in the aggregate; and the equilibrium fixed point is a cubic with closed-form roots, so equilibrium inflows, their elasticity to tail risk, and their response to global conditions are all available analytically.

Three results stand out. First, with positive baseline carry the competitive equilibrium is unique: crowded carry does not by itself manufacture multiple equilibria. When baseline carry is weak and the appreciation feedback is strong, however, the aggregate best-response map admits multiple fixed points, and a sudden stop appears as the disappearance of a self-sustaining boom equilibrium rather than as an imposed binding constraint. This places multiplicity in the aggregate, where it belongs.

Second, the competitive equilibrium is generically inefficient. Atomistic investors ignore both the appreciation their inflows generate for others (a positive pecuniary externality) and the systemic crash risk they create (negative fragility externalities). The

wedge between the competitive and constrained-efficient allocations is closed-form and signs the direction of inefficiency: in inflow booms the market over-invests in tail risk. This connects the model directly to the inefficient-credit-cycle literature of [Lorenzoni \[2008\]](#), [Bianchi \[2011\]](#), and [Mendoza \[2010\]](#), and gives macroprudential policy a Pigouvian foundation.

Third, the policy is most valuable when it keeps the economy out of the region where tail risk becomes steep in inflows. A wedge on foreigners' after-cost return reduces inflows and crash probability, can be made countercyclical in global conditions and local positioning, and can collapse a fragile multiplicity by dampening the appreciation feedback. The margin-style instrument of Appendix B caps positions precisely where endogenous jump risk steepens the variance, making it naturally state-dependent even under a simple rule.

The framework is intentionally partial equilibrium and admits two natural extensions. First, one can endogenize the local-currency return or bond price through market clearing, adding a price channel that would strengthen the amplification mechanism and enlarge the pecuniary externality. Second, one can replace the mean–variance criterion with an explicit utility primitive and a host-country welfare objective, allowing a full welfare-based design of macroprudential rules while preserving the main message: when tail risk is increasing in aggregate positions, managing the cycle in risk-taking and leverage is at least as important as managing the cycle in average flows.

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A Proofs

Proof of Lemma 1. Let the individual investor's objective function be defined as:

$$U_i(f_i; F, g, \tau) = \theta \left(f_i \mu_X(F, g, \tau) - \frac{\kappa}{2} f_i^2 \right) - \frac{\lambda(g)}{2} f_i^2 \sigma_X^2(F, g, \tau)$$

To find the optimal individual position f_i , we first determine the concavity of the objective function with respect to the choice variable. We compute the first and second partial derivatives of U_i with respect to f_i :

The first-order derivative is:

$$\frac{\partial U_i}{\partial f_i} = \theta \mu_X(F, g, \tau) - \theta \kappa f_i - \lambda(g) \sigma_X^2(F, g, \tau) f_i$$

The second-order derivative is:

$$\frac{\partial^2 U_i}{\partial f_i^2} = -\theta \kappa - \lambda(g) \sigma_X^2(F, g, \tau)$$

By assumption, the taste shifter $\theta > 0$, the balance-sheet cost parameter $\kappa > 0$, and the risk aversion coefficient $\lambda(g) > 0$. Furthermore, the variance of the excess return is strictly positive, $\sigma_X^2(F, g, \tau) > 0$. Therefore, it immediately follows that:

$$\frac{\partial^2 U_i}{\partial f_i^2} < 0$$

Because the second derivative is strictly negative globally, the objective function U_i is strictly concave in f_i for any fixed aggregate state (F, g, τ) . This guarantees that any local extremum is the unique global maximum.

Next, we solve for the unconstrained optimal position, denoted as f_i^* , by setting the first-order condition to zero:

$$\theta \mu_X(F, g, \tau) - f_i^* \left(\theta \kappa + \lambda(g) \sigma_X^2(F, g, \tau) \right) = 0$$

Rearranging to isolate f_i^* yields:

$$f_i^* = \frac{\theta \mu_X(F, g, \tau)}{\theta \kappa + \lambda(g) \sigma_X^2(F, g, \tau)}$$

Finally, we must account for the non-negativity constraint $f_i \geq 0$. Because the objective function is a strictly concave parabola opening downwards, the unconstrained maximum f_i^* is the absolute peak of the function.

- If $f_i^* \geq 0$, the constraint is non-binding, and the optimal constrained choice is $f_i = f_i^*$.
- If $f_i^* < 0$, the unconstrained peak lies in the negative domain. Due to the strict concavity of U_i , the function is strictly decreasing for all $f_i > f_i^*$. Consequently, over the feasible domain $[0, \infty)$, the function achieves its maximum at the boundary $f_i = 0$.

Combining these two cases allows us to express the unique best response compactly using the maximum operator:

$$\mathcal{B}(F; g, \tau) = \max \left\{ 0, \frac{\theta \mu_X(F, g, \tau)}{\theta \kappa + \lambda(g) \sigma_X^2(F, g, \tau)} \right\}$$

This concludes the proof. □

Proof of Lemma 2. Recall from Section 2.4 that an interior symmetric competitive equilibrium $F^{\text{CE}} > 0$ must satisfy the fixed-point condition derived from the best response function in Lemma 1:

$$F^{\text{CE}} = \frac{\theta \mu_X(F^{\text{CE}}, g, \tau)}{\theta \kappa + \lambda(g) \sigma_X^2(F^{\text{CE}}, g, \tau)}$$

Rearranging this equation yields the polynomial condition $H(F^{\text{CE}}; g, \tau) = 0$, where the cubic function is defined as:

$$H(F; g, \tau) = a_3 F^3 + a_2 F^2 + a_1 F + a_0$$

First, we evaluate the polynomial at the boundary $F = 0$. Using the coefficients defined in Equation (19):

$$H(0; g, \tau) = a_0 = -\theta A(g, \tau)$$

By assumption, the taste shifter $\theta > 0$ and the baseline expected excess carry $A(g, \tau) > 0$. Therefore, it immediately follows that $H(0; g, \tau) < 0$.

Next, we examine the asymptotic behavior of $H(F; g, \tau)$ as aggregate inflows F grow

arbitrarily large. The leading coefficient of the cubic polynomial is:

$$a_3 = \lambda(g)D(\tau)$$

Since the risk aversion parameter $\lambda(g) > 0$ and the fragility variance scalar $D(\tau) = (1 - \tau)^2 R^2 J^2 \eta > 0$, we have $a_3 > 0$. Because the highest-degree term dominates as $F \rightarrow \infty$, it follows that:

$$\lim_{F \rightarrow \infty} H(F; g, \tau) = +\infty$$

Because $H(F; g, \tau)$ is a polynomial, it is continuous everywhere on the domain $[0, \infty)$. Since $H(0; g, \tau) < 0$ and $H(F; g, \tau) > 0$ for sufficiently large F , the Intermediate Value Theorem guarantees the existence of at least one root $F^* \in (0, \infty)$ such that $H(F^*; g, \tau) = 0$.

Finally, we must verify that this root constitutes a valid competitive equilibrium according to Definition 1. By construction, $H(F^*) = 0$ implies:

$$F^* \left(\theta\kappa + \lambda(g)\sigma_X^2(F^*, g, \tau) \right) = \theta\mu_X(F^*, g, \tau)$$

Because $\theta, \kappa, \lambda > 0$ and $\sigma_X^2 > 0$, the term in the parentheses is strictly positive. Since $F^* > 0$, it must be true that the right-hand side $\theta\mu_X(F^*, g, \tau) > 0$. Consequently, the unconstrained best response evaluated at F^* is strictly positive, meaning the non-negativity constraint in Lemma 1 does not bind:

$$\mathcal{B}(F^*; g, \tau) = \max\{0, F^*\} = F^*$$

Thus, F^* is a fixed point of the best response map. This confirms the existence of at least one interior symmetric competitive equilibrium $F^{\text{CE}} \in (0, \infty)$. $\square$

Proof of Proposition 1. Recall the competitive equilibrium is characterized by the roots of the cubic polynomial $H(F; g, \tau) = a_3 F^3 + a_2 F^2 + a_1 F + a_0 = 0$. From Equation (19), the coefficients are defined as:

$$\begin{aligned} a_3 &= \lambda(g)D(\tau) \\ a_2 &= \theta C(g, \tau) \\ a_1 &= \theta\kappa + \lambda(g)S(g, \tau) - \theta B(\tau) \\ a_0 &= -\theta A(g, \tau) \end{aligned}$$

By the model's baseline parameter restrictions, the risk aversion $\lambda(g) > 0$, the variance scalar $D(\tau) > 0$, the taste shifter $\theta > 0$, and the marginal crash loss $C(g, \tau) > 0$.

Consequently, the two highest-degree coefficients are strictly positive: $a_3 > 0$ and $a_2 > 0$.

We apply Descartes' Rule of Signs to determine the number of positive real roots. We evaluate the sequence of coefficients (a_3, a_2, a_1, a_0) in descending order of degree.

First, consider the case where baseline carry is strictly positive, $A(g, \tau) > 0$. This implies the constant term $a_0 = -\theta A(g, \tau) < 0$. The sign of the linear coefficient a_1 is ambiguous, but we can evaluate both possibilities:

- If $a_1 \geq 0$, the sequence of signs is $(+, +, +, -)$. There is exactly one sign change.
- If $a_1 < 0$, the sequence of signs is $(+, +, -, -)$. There is exactly one sign change.

By Descartes' Rule of Signs, the number of strictly positive real roots is equal to the number of sign changes minus a non-negative even integer. Because there is exactly one sign change in both scenarios, the polynomial $H(F; g, \tau)$ must have exactly one positive real root. Hence, the interior competitive equilibrium is unique.

Second, consider the more general sufficient condition where $a_2 > 0$ and $a_0 \leq 0$. If $a_0 < 0$, the proof of uniqueness holds exactly as above. If $a_0 = 0$ (which occurs when baseline carry $A(g, \tau) = 0$), the polynomial factors as:

$$H(F; g, \tau) = F(a_3 F^2 + a_2 F + a_1) = 0$$

One root is trivially $F = 0$. The remaining positive roots must satisfy the quadratic equation $a_3 F^2 + a_2 F + a_1 = 0$. We again apply Descartes' Rule of Signs to the sequence (a_3, a_2, a_1) , knowing $a_3 > 0$ and $a_2 > 0$:

- If $a_1 \geq 0$, the signs are $(+, +, +)$. There are zero sign changes, meaning there are no positive real roots.
- If $a_1 < 0$, the signs are $(+, +, -)$. There is exactly one sign change, meaning there is exactly one positive real root.

Thus, when $a_0 = 0$, the system yields either zero or one strictly positive root. Combining this with the $a_0 < 0$ case proves that whenever $a_2 > 0$ and $a_0 \leq 0$ (which holds naturally when $C > 0$ and $A \geq 0$), there is at most one positive interior equilibrium.

□

Proof of Proposition 2. To find the roots of the cubic equation $H(F; g, \tau) = a_3 F^3 + a_2 F^2 + a_1 F + a_0 = 0$, we first apply the Tschirnhaus transformation to eliminate the

quadratic term. We substitute $F = y - \frac{a_2}{3a_3}$ into the equation. Expanding the terms yields:

$$a_3 \left(y^3 - 3y^2 \frac{a_2}{3a_3} + 3y \left(\frac{a_2}{3a_3} \right)^2 - \left(\frac{a_2}{3a_3} \right)^3 \right) + a_2 \left(y^2 - 2y \frac{a_2}{3a_3} + \left(\frac{a_2}{3a_3} \right)^2 \right) + a_1 \left(y - \frac{a_2}{3a_3} \right) + a_0 = 0$$

Notice that the coefficient for y^2 collapses perfectly: $a_3(-a_2/a_3) + a_2 = 0$. Dividing the entire equation by the strictly positive leading coefficient a_3 leaves the monic depressed cubic:

$$y^3 + py + q = 0$$

where gathering the constants produces exactly $p = \frac{3a_3a_1 - a_2^2}{3a_3^2}$ and $q = \frac{2a_2^3 - 9a_3a_2a_1 + 27a_3^2a_0}{27a_3^3}$.

The nature of the roots of this depressed cubic is governed by its discriminant, $\Delta = \left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3$.

Case 1: $\Delta > 0$ (One real root). By Cardano's method, we let $y = u + v$, which implies $y^3 = u^3 + v^3 + 3uv(u + v) = u^3 + v^3 + 3uvy$. Equating this term-by-term to $y^3 + py + q = 0$ yields the system $3uv = -p$ and $u^3 + v^3 = -q$. Cubing the first equation gives $u^3v^3 = -p^3/27$. Thus, u^3 and v^3 are the two roots of the quadratic equation $Z^2 + qZ - p^3/27 = 0$. Applying the quadratic formula:

$$Z = -\frac{q}{2} \pm \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3} = -\frac{q}{2} \pm \sqrt{\Delta}$$

Taking the cube roots isolates u and v , yielding the single real root:

$$y^* = u + v = \sqrt[3]{-\frac{q}{2} + \sqrt{\Delta}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\Delta}}$$

The corresponding aggregate inflow is $F^{\text{CE}} = y^* - \frac{a_2}{3a_3}$.

Case 2: $\Delta < 0$ (Three real roots). This is the *casus irreducibilis*, where Cardano's formula requires the cube root of a complex number. Since $\Delta < 0$, it is algebraically necessary that $p < 0$. We use François Viète's trigonometric substitution. Recall the identity $4\cos^3(\theta) - 3\cos(\theta) = \cos(3\theta)$. We substitute $y = u\cos(\theta)$ into the depressed cubic:

$$u^3\cos^3(\theta) + pu\cos(\theta) + q = 0 \implies 4\cos^3(\theta) + \frac{4p}{u^2}\cos(\theta) + \frac{4q}{u^3} = 0$$

Matching coefficients with the identity, we set $\frac{4p}{u^2} = -3$, which yields $u = 2\sqrt{-\frac{p}{3}}$. We

then match the constant term:

$$\frac{4q}{u^3} = -\cos(3\theta) \implies \cos(3\theta) = -\frac{4q}{8(-\frac{p}{3})^{3/2}} = \frac{3q}{2p} \sqrt{-\frac{3}{p}}$$

Because $\Delta < 0$, the argument of the arccosine is strictly bounded between -1 and 1. Solving for θ provides three distinct angles reflecting the periodic nature of the cosine function:

$$\theta_k = \frac{1}{3} \arccos \left(\frac{3q}{2p} \sqrt{-\frac{3}{p}} \right) - \frac{2\pi k}{3}, \quad \text{for } k = 0, 1, 2$$

This yields three distinct real roots $y_k = 2\sqrt{-\frac{p}{3}} \cos(\theta_k)$, which map to $F_k = y_k - \frac{a_2}{3a_3}$.

Case 3: $\Delta = 0$ (Repeated real roots). The algebraic solutions transition seamlessly. If $\Delta = 0$ and $p = q = 0$, then $y = 0$ is a root of multiplicity three. Otherwise, it produces a single real root $y = \frac{3q}{p}$ and a double real root $y = -\frac{3q}{2p}$.

Finally, because all baseline parameters a_0, a_1, a_2, a_3 are real, the shift constant $a_2/(3a_3)$ is purely real. Therefore, any real root y maps to a strictly real aggregate inflow F . Since position sizes cannot be negative, the economically meaningful equilibria are restricted to the non-negative real roots $F \geq 0$. $\square$

Proof of Proposition 3. We define the planner's constrained-efficient objective function over the aggregate inflow F as:

$$W(F) = \theta \left(F \mu_X(F) - \frac{\kappa}{2} F^2 \right) - \frac{\lambda(g)}{2} F^2 \sigma_X^2(F)$$

To find the marginal social value of an additional unit of inflow, we differentiate $W(F)$ with respect to F . Using the product rule to carefully separate the direct effect of more inflows from the indirect effect on prices and risk, we obtain:

$$W_F(F) = \underbrace{\left[\theta \mu_X(F) - \theta \kappa F - \lambda(g) F \sigma_X^2(F) \right]}_{\text{Direct effect}} + \underbrace{\left[\theta F \mu'_X(F) - \frac{\lambda(g)}{2} F^2 (\sigma_X^2)'(F) \right]}_{\text{Externality effect}}$$

Notice that the first bracketed term is exactly the marginal private value of inflows evaluated at the symmetric allocation $f_i = F$:

$$\text{MV}^{\text{priv}}(F) = \theta \mu_X(F) - \theta \kappa F - \lambda(g) F \sigma_X^2(F)$$

This is the object the atomistic investor sets to zero (as shown in Lemma 1 and the

equilibrium condition in Equation (18)).

The wedge between the social and private marginal values is therefore precisely the second bracketed term, which captures how an extra unit of aggregate inflow alters the expected return and variance for the entire stock of existing inflows:

$$\delta(F) \equiv W_F(F) - \text{MV}^{\text{priv}}(F) = \theta F \mu'_X(F) - \frac{\lambda(g)}{2} F^2 (\sigma_X^2)'(F)$$

We now substitute the explicit functional forms from Equation (15):

$$\begin{aligned} \mu_X(F) &= A + BF - CF^2 \implies \mu'_X(F) = B - 2CF \\ \sigma_X^2(F) &= S + DF^2 \implies (\sigma_X^2)'(F) = 2DF \end{aligned}$$

Plugging these derivatives into the wedge equation yields:

$$\delta(F) = \theta F(B - 2CF) - \frac{\lambda(g)}{2} F^2 (2DF)$$

Expanding and collecting the terms, we arrive at the closed-form expression for the externality wedge:

$$\delta(F) = \theta BF - 2\theta CF^2 - \lambda(g)DF^3$$

Factoring out F yields the final result:

$$\delta(F) = F(\theta B - 2\theta CF - \lambda(g)DF^2)$$

This completes the proof. □

Proof of Proposition 4. From Proposition 3, the relationship between the marginal social value and the marginal private value of aggregate inflows is given by:

$$W_F(F) = \text{MV}^{\text{priv}}(F) + \delta(F)$$

By Definition 1 and Lemma 1, the interior competitive equilibrium F^{CE} is characterized by the first-order condition of the atomistic investor, meaning $\text{MV}^{\text{priv}}(F^{\text{CE}}) = 0$. Evaluating the marginal social value at the competitive equilibrium therefore isolates the externality wedge:

$$W_F(F^{\text{CE}}) = \delta(F^{\text{CE}}) = F^{\text{CE}}(\theta B - 2\theta CF^{\text{CE}} - \lambda(g)D(F^{\text{CE}})^2)$$

To determine the sign of the wedge, we define the quadratic function representing the

bracketed terms, factoring out a negative sign for algebraic convenience:

$$q(F) \equiv \lambda(g)DF^2 + 2\theta CF - \theta B$$

This allows us to write the wedge as $\delta(F^{\text{CE}}) = -F^{\text{CE}} \cdot q(F^{\text{CE}})$.

We analyze the roots of $q(F) = 0$. The y -intercept is $q(0) = -\theta B$. Since $\theta > 0$ and $B > 0$ (assuming a strictly positive portfolio-balance channel $\chi > 0$), $q(0) < 0$. Because the leading coefficient $\lambda(g)D > 0$, the parabola opens upward. By the Intermediate Value Theorem, $q(F)$ must cross the horizontal axis exactly once in the positive domain. We find this unique positive root, $\hat{F}$, using the quadratic formula:

$$\hat{F} = \frac{-2\theta C + \sqrt{(2\theta C)^2 - 4(\lambda(g)D)(-\theta B)}}{2\lambda(g)D} = \frac{-\theta C + \sqrt{\theta^2 C^2 + \lambda(g)D \theta B}}{\lambda(g)D}$$

Because $q(F)$ is an upward-opening parabola with a negative y -intercept, it follows that:

- $q(F) < 0$ for all $F \in [0, \hat{F})$
- $q(F) > 0$ for all $F \in (\hat{F}, \infty)$

We now map this behavior to the sign of the marginal social value at the competitive equilibrium and the resulting direction of inefficiency. Recall that the planner's objective function $W(F)$ in Equation (24) is a polynomial of degree four. Its leading term is $-\frac{\lambda(g)D}{2}F^4$. Because the coefficient is strictly negative, $\lim_{F \rightarrow \infty} W(F) = -\infty$, meaning $W(F)$ is coercive and attains a global maximum at some interior constrained-efficient allocation F^{SP} .

Case 1: Over-investment ($F^{\text{CE}} > \hat{F}$). If the competitive inflow exceeds the threshold, then $q(F^{\text{CE}}) > 0$. Since $F^{\text{CE}} > 0$, the wedge is strictly negative: $\delta(F^{\text{CE}}) = -F^{\text{CE}} \cdot q(F^{\text{CE}}) < 0$. Consequently, $W_F(F^{\text{CE}}) < 0$. The marginal social value of an additional unit of inflow is negative at the competitive equilibrium, implying that the planner can strictly increase welfare by reducing aggregate inflows. The fragility externalities dominate the appreciation externality, leading the competitive market to over-invest: $F^{\text{CE}} > F^{\text{SP}}$.

Case 2: Under-investment ($F^{\text{CE}} < \hat{F}$). If the competitive inflow is below the threshold, then $q(F^{\text{CE}}) < 0$. This forces the wedge to be strictly positive: $\delta(F^{\text{CE}}) = -F^{\text{CE}} \cdot q(F^{\text{CE}}) > 0$. Consequently, $W_F(F^{\text{CE}}) > 0$. The marginal social value of inflows is positive, implying the planner can strictly increase welfare by encouraging more inflows. Here, the positive pecuniary appreciation externality dominates the negative fragility externalities, causing the competitive market to under-invest: $F^{\text{CE}} < F^{\text{SP}}$. $\square$

Proof of Proposition 5. The proof proceeds in three steps: establishing the target efficient allocation, analyzing the boundaries of the competitive equilibrium with respect to the wedge, and applying the Intermediate Value Theorem to find the optimal policy.

Step 1: The constrained-efficient target. We assume the policy wedge revenue is rebated lump-sum, so the planner evaluates welfare using gross (pre-transfer) returns, setting $\tau = 0$. The constrained-efficient allocation is $F^{\text{SP}}(g) = \arg \max_F W(F; g, 0)$. Recall from Equation (24) that $W(F; g, 0)$ is a degree-four polynomial with a strictly negative leading term, $-\frac{\lambda(g)D(0)}{2}F^4$. Thus, $W(F)$ is coercive ($\lim_{F \rightarrow \infty} W(F) = -\infty$), guaranteeing a global maximum. Furthermore, evaluating the marginal social value at zero gives $W_F(0; g, 0) = \theta A(g, 0)$. Assuming baseline carry is positive ($A > 0$), this marginal value is strictly positive. Consequently, the global maximum cannot be at the boundary zero, ensuring the target allocation is strictly interior: $F^{\text{SP}}(g) > 0$.

Step 2: The competitive equilibrium bounds. By the hypothesis of the proposition, the unregulated market over-invests due to dominant fragility externalities: $F^{\text{CE}}(g, 0) > F^{\text{SP}}(g)$. Now consider the behavior of the competitive equilibrium $F^{\text{CE}}(g, \tau)$ as the wedge τ increases toward 1 (a 100% tax on the local return). Recall the baseline expected excess carry from Equation (10):

$$A(g, \tau) = (1 - \tau)R(1 + \mu_e(g) - J\bar{\pi}(g)) - R^*(g)$$

As $\tau \rightarrow 1$, the first term vanishes, leaving $\lim_{\tau \rightarrow 1} A(g, \tau) = -R^*(g) < 0$. Because the baseline carry becomes strictly negative, no investor will hold the asset even at zero aggregate positioning. Consequently, the interior competitive root is eliminated, forcing the equilibrium to the boundary: $\lim_{\tau \rightarrow 1} F^{\text{CE}}(g, \tau) = 0$.

Step 3: Implementation via the Intermediate Value Theorem. By Proposition 7, the competitive equilibrium $F^{\text{CE}}(g, \tau)$ is a continuous and strictly decreasing function of τ on the interior region. We have established that the function maps the interval $[0, 1)$ continuously from $F^{\text{CE}}(g, 0)$ down to 0. Since the target allocation lies strictly within these bounds, $0 < F^{\text{SP}}(g) < F^{\text{CE}}(g, 0)$, the Intermediate Value Theorem guarantees the existence of at least one wedge $\tau^* \in (0, 1)$ such that:

$$F^{\text{CE}}(g, \tau^*) = F^{\text{SP}}(g)$$

Furthermore, because $F^{\text{CE}}(g, \tau)$ is strictly decreasing in τ , this implementing wedge τ^* is unique.

Economically, because the wedge revenue is treated as a pure transfer, the social objective function $W(F)$ itself is unaffected by τ . The wedge strictly alters the atomistic investors' private first-order condition: lowering their perceived marginal return just

enough to internalize the negative fragility externality they otherwise ignore. Thus, τ^* serves as a perfect Pigouvian tax, decentralizing the constrained-efficient allocation. $\square$

Proof of Theorem 1. The proof proceeds in four parts: verifying the boundary equilibrium, establishing the existence of two interior equilibria, analyzing their stability, and characterizing the sudden stop bifurcation.

Step 1: The no-inflow equilibrium. By hypothesis, baseline carry is unattractive at zero positioning: $A(g, \tau) < 0$. At $F = 0$, the expected excess return is $\mu_X(0) = A < 0$. Evaluating the best-response function from Lemma 1 yields:

$$\mathcal{B}(0; g, \tau) = \max \left\{ 0, \frac{\theta A}{\theta \kappa + \lambda(g)S(g, \tau)} \right\} = 0$$

Because $\mathcal{B}(0) = 0$, the boundary $F = 0$ constitutes a valid competitive equilibrium.

Step 2: Existence of the interior boom equilibria. Interior equilibria satisfy the cubic condition $H(F) = a_3 F^3 + a_2 F^2 + a_1 F + a_0 = 0$. Because $A < 0$, the y -intercept is strictly positive: $H(0) = a_0 = -\theta A > 0$. The leading coefficient $a_3 = \lambda(g)D > 0$, so $\lim_{F \rightarrow \infty} H(F) = +\infty$. To find turning points, we analyze the derivative:

$$H'(F) = 3a_3 F^2 + 2a_2 F + a_1$$

Condition (28) explicitly states that the pecuniary appreciation feedback dominates the balance-sheet and baseline variance costs, meaning $a_1 = \theta \kappa + \lambda(g)S - \theta B < 0$. Since $H'(F)$ is an upward-opening parabola with a negative y -intercept ($H'(0) = a_1 < 0$), it has exactly one positive root, denoted F_m . Consequently, $H(F)$ attains a unique local minimum at F_m in the positive domain. If the appreciation feedback is sufficiently strong, it drives the polynomial deep enough such that $H(F_m) < 0$. Because $H(0) > 0$, $H(F_m) < 0$, and $\lim_{F \rightarrow \infty} H(F) > 0$, the Intermediate Value Theorem dictates that $H(F)$ must cross the horizontal axis exactly twice in $(0, \infty)$, yielding two positive roots: $0 < F_L < F_m < F_H$.

Step 3: Best-response stability. We assess stability using the iterative mapping $F_{t+1} = \mathcal{B}(F_t)$. From the definition of $H(F)$, for any $F > 0$ where the interior unconstrained best response is positive, we can factor the equilibrium condition:

$$H(F) = F(\theta \kappa + \lambda(g)\sigma_X^2(F)) - \theta \mu_X(F) = (\theta \kappa + \lambda(g)\sigma_X^2(F)) [F - \mathcal{B}(F)]$$

Because the risk and cost penalties are strictly positive, the sign of $F - \mathcal{B}(F)$ perfectly matches the sign of $H(F)$.

- **At $F = 0$:** For $F \in (0, F_L)$, $H(F) > 0$, which implies $F > \mathcal{B}(F)$. The market size naturally contracts back to zero. Thus, $F = 0$ is locally stable, with a basin of attraction $[0, F_L)$.
- **At F_L (The tipping point):** $H(F)$ crosses from positive to negative, meaning $H'(F_L) < 0$. For F slightly below F_L , the system contracts toward 0. For F slightly above F_L , $H(F) < 0 \implies \mathcal{B}(F) > F$, so the system expands away from F_L . Thus, F_L is an unstable equilibrium.
- **At F_H (The boom):** $H(F)$ crosses from negative to positive, meaning $H'(F_H) > 0$. For $F \in (F_L, F_H)$, $H(F) < 0 \implies \mathcal{B}(F) > F$, pushing inflows up toward F_H . For $F > F_H$, $H(F) > 0 \implies \mathcal{B}(F) < F$, pulling inflows down toward F_H . Thus, F_H is locally stable.

Step 4: Sudden stops via bifurcation. Suppose the economy is resting at the high-inflow boom equilibrium F_H . A deterioration in global conditions (a decrease in g) or a tightening of the macroprudential wedge (an increase in τ) directly reduces the baseline carry A and suppresses the appreciation multiplier B . Algebraically, these changes strictly increase the coefficients a_0 and a_1 , shifting the entire curve $H(F)$ vertically upward. At a critical parameter threshold, the local minimum $H(F_m)$ is lifted above zero. At this point, the roots F_L and F_H annihilate one another in a saddle-node bifurcation. Once $H(F_m) > 0$, we have $H(F) > 0$ for all $F > 0$. This implies $\mathcal{B}(F) < F$ globally. The aggregate best-response map no longer crosses the 45° line in the positive quadrant. The market unravels, and inflows contract continuously until they collapse entirely to the unique remaining equilibrium at $F = 0$. This discontinuous jump models a self-fulfilling sudden stop. $\square$

Proof of Corollary 1. The corollary investigates the regime structure when the macroprudential wedge τ is tightened sufficiently to violate the strong feedback condition from Theorem 1. Specifically, suppose τ is raised such that the portfolio-balance appreciation feedback is fully dampened by balance-sheet and variance costs:

$$\theta B(\tau) \leq \theta \kappa + \lambda(g)S(g, \tau)$$

Recall the linear coefficient of the equilibrium cubic $H(F)$ from Equation (19) is $a_1 = \theta \kappa + \lambda(g)S(g, \tau) - \theta B(\tau)$. Under the condition above, this coefficient becomes non-negative: $a_1 \geq 0$.

To determine the number of possible equilibria, we analyze the slope of the cubic

polynomial $H(F)$ for all $F > 0$. The derivative is:

$$H'(F) = 3a_3F^2 + 2a_2F + a_1$$

By the model's baseline assumptions, the cubic and quadratic coefficients are strictly positive: $a_3 = \lambda(g)D(\tau) > 0$ and $a_2 = \theta C(g, \tau) > 0$. Since $a_1 \geq 0$, every term in the derivative is strictly positive for any $F > 0$. Consequently:

$$H'(F) > 0 \quad \text{for all } F > 0$$

This implies that $H(F)$ is a strictly monotonically increasing function on the positive real line. Geometrically, the polynomial can no longer feature a local minimum or "dip," meaning it can cross the horizontal axis at most once. The pair of interior roots (F_L, F_H) from Theorem 1 is mathematically impossible.

The exact nature of the unique remaining equilibrium depends entirely on the y -intercept, $H(0) = -\theta A(g, \tau)$:

- **Case 1: Positive baseline carry ($A > 0$).** If baseline carry is attractive, $H(0) < 0$. Because $H(F)$ is continuous, strictly increasing, and $\lim_{F \rightarrow \infty} H(F) = \infty$, it must cross the axis exactly once. The system admits a unique interior competitive equilibrium $F^{\text{CE}} > 0$.
- **Case 2: Non-positive baseline carry ($A \leq 0$).** If baseline carry is unattractive, $H(0) \geq 0$. Because $H(F)$ is strictly increasing, $H(F) > 0$ for all $F > 0$. There are no interior roots. The unconstrained best response is always negative, so the boundary constraint binds everywhere, leaving $F = 0$ as the globally unique equilibrium.

In neither case can a high-inflow boom equilibrium coexist with a no-inflow equilibrium. By shrinking the appreciation multiplier $B(\tau) = (1 - \tau)R\chi$, the macroprudential wedge physically flattens the feedback loop that generates self-fulfilling expectations, thereby completely ruling out the discontinuous switching dynamics (sudden stops). $\square$

Proof of Lemma 3. The proof proceeds in two steps. First, we establish the limiting behavior of the risk price $\rho(F)$ as aggregate inflows grow arbitrarily large. Second, we prove strict monotonicity for large F by analyzing the sign of its derivative.

Step 1: Asymptotic limit of the risk price. From Equation (29), the aggregate risk

price is defined as:

$$\rho(F) = \frac{\mu_X(F)}{\sigma_X(F)} = \frac{A + BF - CF^2}{\sqrt{S + DF^2}}$$

To evaluate the limit as $F \rightarrow \infty$, we factor out F from both the numerator and the denominator. Since $F > 0$, we can write $\sqrt{S + DF^2} = F\sqrt{\frac{S}{F^2} + D}$. Thus:

$$\rho(F) = \frac{F\left(\frac{A}{F} + B - CF\right)}{F\sqrt{\frac{S}{F^2} + D}} = \frac{\frac{A}{F} + B - CF}{\sqrt{\frac{S}{F^2} + D}}$$

As $F \rightarrow \infty$, the terms A/F and S/F^2 vanish to zero. Because $C > 0$, the numerator diverges to $-\infty$. Because $D > 0$, the denominator converges to the strictly positive constant $\sqrt{D}$. Therefore:

$$\lim_{F \rightarrow \infty} \rho(F) = -\infty$$

This confirms that at extremely high aggregate inflows, the endogenous crash losses strictly dominate any appreciation benefits, driving the expected risk-adjusted payoff deeply negative.

Step 2: Strict monotonicity for high inflows. To determine where the risk price is decreasing, we differentiate $\rho(F)$ with respect to F . Applying the standard quotient rule yields:

$$\rho'(F) = \frac{\mu'_X(F)\sigma_X(F) - \mu_X(F)\sigma'_X(F)}{\sigma_X^2(F)}$$

Recall the derivatives of our components:

$$\begin{aligned}\mu'_X(F) &= B - 2CF \\ \sigma'_X(F) &= \frac{d}{dF}(S + DF^2)^{1/2} = \frac{DF}{\sqrt{S + DF^2}}\end{aligned}$$

Substituting these into the quotient rule:

$$\rho'(F) = \frac{(B - 2CF)\sqrt{S + DF^2} - (A + BF - CF^2)\frac{DF}{\sqrt{S + DF^2}}}{S + DF^2}$$

To simplify, we multiply the numerator and denominator by $\sqrt{S + DF^2}$:

$$\rho'(F) = \frac{(B - 2CF)(S + DF^2) - (A + BF - CF^2)DF}{(S + DF^2)^{3/2}}$$

Because the denominator $(S + DF^2)^{3/2}$ is strictly positive for all F , the sign of $\rho'(F)$ depends entirely on the sign of the numerator, which we define as $\mathcal{N}(F)$. Expanding

the terms in $\mathcal{N}(F)$ gives:

$$\begin{aligned}(B - 2CF)(S + DF^2) &= BS + BDF^2 - 2CSF - 2CDF^3 \\ -(A + BF - CF^2)DF &= -ADF - BDF^2 + CDF^3\end{aligned}$$

Adding these expansions together, the BDF^2 terms cancel cleanly:

$$\mathcal{N}(F) = -CDF^3 - (2CS + AD)F + BS$$

This establishes that the numerator $\mathcal{N}(F)$ is a cubic polynomial in F . Because $C > 0$ and $D > 0$, the leading coefficient $-CD$ is strictly negative. Therefore, as $F \rightarrow \infty$, the cubic function dictates that $\lim_{F \rightarrow \infty} \mathcal{N}(F) = -\infty$.

By the definition of the limit, there must exist some threshold $\bar{F}_\rho > 0$ such that for all $F > \bar{F}_\rho$, the numerator $\mathcal{N}(F) < 0$. Consequently, $\rho'(F) < 0$ for all $F > \bar{F}_\rho$. Thus, the risk price is strictly decreasing in the high-inflow region. $\square$

Proof of Proposition 6. By definition, the log-log elasticity of the tail risk probability $\mathcal{T}$ with respect to aggregate inflows F measures the percentage change in crash risk for a 1% change in aggregate positioning. It is computed as:

$$\varepsilon_{\mathcal{T},F}(F, g) \equiv \frac{\partial \log \mathcal{T}}{\partial \log F} = \frac{\partial \mathcal{T}}{\partial F} \frac{F}{\mathcal{T}}$$

Recall the functional form of the crash probability from Equation (3):

$$\mathcal{T}(F, g) = \pi(F, g) = \bar{\pi}(g) + \eta F^2$$

Differentiating this with respect to F yields the marginal increase in crash probability:

$$\frac{\partial \mathcal{T}}{\partial F} = 2\eta F$$

Substituting this derivative and the definition of $\mathcal{T}$ back into the elasticity formula gives:

$$\varepsilon_{\mathcal{T},F}(F, g) = (2\eta F) \frac{F}{\bar{\pi}(g) + \eta F^2} = \frac{2\eta F^2}{\bar{\pi}(g) + \eta F^2}$$

This establishes the exact formula in Equation (30).

Next, we consider the asymptotic behavior in a crowded market where endogenous fragility overwhelmingly dominates the baseline crash risk, meaning $\eta F^2 \gg \bar{\pi}(g)$. To evaluate the limit formally as aggregate inflows grow large ($F \rightarrow \infty$), we divide both

the numerator and the denominator by the endogenous risk term ηF^2 :

$$\varepsilon_{\mathcal{T},F}(F, g) = \frac{\frac{2\eta F^2}{\eta F^2}}{\frac{\bar{\pi}(g)}{\eta F^2} + \frac{\eta F^2}{\eta F^2}} = \frac{2}{\frac{\bar{\pi}(g)}{\eta F^2} + 1}$$

Because the baseline crash probability $\bar{\pi}(g)$ is a fixed parameter for any given global state g , the term $\frac{\bar{\pi}(g)}{\eta F^2}$ vanishes to zero as $F \rightarrow \infty$. Evaluating the limit yields:

$$\lim_{F \rightarrow \infty} \varepsilon_{\mathcal{T},F}(F, g) = \frac{2}{0 + 1} = 2$$

Economically, because fragility is quadratic in inflows, the elasticity converges asymptotically to 2. In high-inflow states, every 1% increase in the aggregate position scale drives a proportional 2% increase in the likelihood of a sudden exchange rate collapse.

□

Proof of Proposition 7. We define the implicit function $\Psi(F; \tau)$ to represent the marginal private value of inflows, exactly as it appears in the competitive equilibrium condition $\text{MV}^{\text{priv}}(F) = 0$. Using the structural definitions from Equations (7) and (9), we write the condition explicitly in terms of the wedge τ :

$$\Psi(F; \tau) \equiv \theta \left[(1 - \tau) R m(F) - R^*(g) - \kappa F \right] - \lambda(g) F (1 - \tau)^2 R^2 s(F) = 0$$

where we have bundled the terms that depend purely on F and g :

$$\begin{aligned} m(F) &= 1 + \mu_e(g) + \chi F - J\pi(F, g) \\ s(F) &= \sigma_\varepsilon^2(g) + J^2\pi(F, g) \end{aligned}$$

By the Implicit Function Theorem, the sensitivity of the competitive equilibrium to the macroprudential wedge is:

$$\frac{\partial F^{\text{CE}}}{\partial \tau} = - \frac{\Psi_\tau(F^{\text{CE}}; \tau)}{\Psi_F(F^{\text{CE}}; \tau)}$$

First, consider the denominator Ψ_F . Because $\Psi(F; \tau)$ is identically the negative of the equilibrium cubic $H(F; g, \tau)$ defined in Equation (18), we know that $\Psi_F = -H_F$. By the hypothesis of the proposition, we are evaluating a unique, stable interior competitive equilibrium where the cubic crosses zero from below, meaning $H_F(F^{\text{CE}}) > 0$. Consequently, the denominator $\Psi_F(F^{\text{CE}}; \tau) < 0$.

Next, we evaluate the numerator by differentiating Ψ with respect to τ :

$$\Psi_\tau = -\theta R m(F) + 2\lambda(g)F(1-\tau)R^2s(F)$$

To sign this expression, we use the equilibrium condition $\Psi = 0$ to substitute out the variance term. Rearranging $\Psi = 0$ gives:

$$\lambda(g)F(1-\tau)^2R^2s(F) = \theta \left[(1-\tau)R m(F) - R^*(g) - \kappa F \right]$$

Dividing both sides by $(1-\tau)$ yields an expression for the exact term we need:

$$\lambda(g)F(1-\tau)R^2s(F) = \frac{\theta}{1-\tau} \left[(1-\tau)R m(F) - R^*(g) - \kappa F \right]$$

Substitute this back into Ψ_τ :

$$\begin{aligned} \Psi_\tau &= -\theta R m(F) + \frac{2\theta}{1-\tau} \left[(1-\tau)R m(F) - R^*(g) - \kappa F \right] \\ &= \frac{\theta}{1-\tau} \left[-(1-\tau)R m(F) + 2(1-\tau)R m(F) - 2R^*(g) - 2\kappa F \right] \\ &= \frac{\theta}{1-\tau} \left[(1-\tau)R m(F) - 2R^*(g) - 2\kappa F \right] \end{aligned}$$

To interpret this economically, we again use the equilibrium condition $\Psi = 0$ to express the expected gross local return:

$$(1-\tau)R m(F) = R^*(g) + \kappa F + \underbrace{\frac{\lambda(g)}{\theta}(1-\tau)^2R^2Fs(F)}_{\text{Risk Premium} > 0}$$

Substituting this into our simplified Ψ_τ :

$$\Psi_\tau = \frac{\theta}{1-\tau} \left[\text{Risk Premium} - R^*(g) - \kappa F \right]$$

For any plausibly calibrated economy, the baseline global gross funding rate $R^*(g)$ is greater than 1 (e.g., 1.05). The equilibrium risk premium on currency carry trades is a fractional return on the order of a few percent (e.g., 0.03), and $\kappa F \geq 0$. Therefore, $\text{Risk Premium} \ll R^*(g)$, rendering the bracketed term strictly negative. Thus, $\Psi_\tau < 0$.

Combining the signs via the Implicit Function Theorem yields:

$$\frac{\partial F^{\text{CE}}}{\partial \tau} = -\frac{(-)}{(-)} < 0$$

Increasing the macroprudential tax strictly reduces equilibrium aggregate inflows.

Finally, recall that the tail risk probability is $\mathcal{T}(F, g) = \bar{\pi}(g) + \eta(F^{\text{CE}})^2$. Applying the chain rule gives:

$$\frac{\partial \mathcal{T}}{\partial \tau} = 2\eta F^{\text{CE}} \left(\frac{\partial F^{\text{CE}}}{\partial \tau} \right) < 0$$

Because inflows fall, the endogenous crash probability strictly decreases as well. $\square$

Proof of Proposition 8. We evaluate the sensitivity of the competitive equilibrium to the global financial cycle using the Implicit Function Theorem. Let the implicit function $\Psi(F; g)$ represent the marginal private value of inflows from the competitive equilibrium condition:

$$\Psi(F; g) \equiv \theta \mu_X(F, g, \tau) - \theta \kappa F - \lambda(g) F \sigma_X^2(F, g, \tau) = 0$$

By the Implicit Function Theorem, the response of aggregate inflows to an improvement in global conditions g is given by:

$$\frac{\partial F^{\text{CE}}}{\partial g} = - \frac{\Psi_g(F^{\text{CE}}; g)}{\Psi_F(F^{\text{CE}}; g)}$$

As established in the proof of Proposition 7, the denominator $\Psi_F = -H_F$. At the unique, stable interior competitive equilibrium, the equilibrium cubic crosses zero from below, meaning $H_F(F^{\text{CE}}) > 0$. This implies $\Psi_F < 0$. Consequently, the sign of $\frac{\partial F^{\text{CE}}}{\partial g}$ perfectly matches the sign of the partial derivative Ψ_g .

We differentiate $\Psi(F; g)$ with respect to g , holding F constant:

$$\Psi_g = \theta \frac{\partial \mu_X}{\partial g} - \lambda'(g) F \sigma_X^2 - \lambda(g) F \frac{\partial \sigma_X^2}{\partial g}$$

We analyze the sign of this expression term-by-term using the structural definitions in Equation (15) and the monotonicities specified in Assumption 3.

First term (Expected return effect): Recall that $\mu_X(F, g, \tau) = A(g, \tau) + B(\tau)F - C(g, \tau)F^2$. Note that the portfolio-balance coefficient $B(\tau)$ and the marginal crash loss $C(g, \tau) = (1 - \tau)RJ\eta$ do not depend on g . Thus, $\frac{\partial \mu_X}{\partial g} = \frac{\partial A}{\partial g}$. Expanding the baseline carry $A(g, \tau)$ yields:

$$\frac{\partial A}{\partial g} = (1 - \tau)R(\mu'_e(g) - J\bar{\pi}'(g)) - (R^*)'(g)$$

By Assumption 3, easier global conditions increase baseline appreciation ($\mu'_e \geq 0$), reduce baseline crash probability ($\bar{\pi}' < 0$), and lower funding costs ($(R^*)' < 0$). Thus, $\frac{\partial A}{\partial g} > 0$ (which perfectly satisfies the premise of the proposition). This makes the first

term strictly positive: $\theta \frac{\partial A}{\partial g} > 0$.

Second term (Risk appetite effect): Assumption 3 states that easier global conditions lower the effective risk-aversion coefficient, meaning $\lambda'(g) < 0$. Since aggregate inflows $F > 0$ and total variance $\sigma_X^2 > 0$, the negative sign cancels out, yielding a strictly positive contribution: $-\lambda'(g)F\sigma_X^2 > 0$.

Third term (Risk quantity effect): Recall that $\sigma_X^2(F, g, \tau) = S(g, \tau) + D(\tau)F^2$. Since $D(\tau)$ is independent of g , we have $\frac{\partial \sigma_X^2}{\partial g} = \frac{\partial S}{\partial g}$. Expanding the baseline variance $S(g, \tau)$:

$$\frac{\partial S}{\partial g} = (1 - \tau)^2 R^2 \left(\frac{\partial \sigma_\varepsilon^2(g)}{\partial g} + J^2 \bar{\pi}'(g) \right)$$

By Assumption 3, both normal-times volatility and baseline crash probability decline with g , meaning $\frac{\partial \sigma_\varepsilon^2(g)}{\partial g} \leq 0$ and $\bar{\pi}' < 0$. Thus, $\frac{\partial S}{\partial g} < 0$. Because $\lambda(g) > 0$ and $F > 0$, subtracting this term yields a strictly positive result: $-\lambda(g)F \frac{\partial S}{\partial g} > 0$.

Summing these three strictly positive components confirms that the marginal private value of inflows strictly increases with easier global conditions:

$$\Psi_g = \underbrace{\theta \frac{\partial A}{\partial g}}_{>0} + \underbrace{(-\lambda'(g))F\sigma_X^2}_{>0} + \underbrace{\lambda(g)F \left(-\frac{\partial S}{\partial g} \right)}_{>0} > 0$$

Returning to the Implicit Function Theorem:

$$\frac{\partial F^{\text{CE}}}{\partial g} = -\frac{(+)}{(-)} > 0$$

Therefore, an improvement in the global financial cycle unambiguously drives an expansion in aggregate carry trade inflows. $\square$

B A VaR-Style Margin Constraint

Some macroprudential toolkits operate through margins rather than direct taxes. A Value-at-Risk constraint on the position takes the form

$$\text{VaR}_\alpha(f) \leq mW, \quad (33)$$

where $\alpha \in (0, 1)$ is the tail confidence level, $m \in (0, 1)$ is margin tightness set by the regulator (lower m is tighter), and $W > 0$ is investor wealth or intermediation capacity. Under the normal-approximation $\text{VaR}_\alpha(f) \approx z_\alpha f \sigma_X(F, g, \tau)$ with $z_\alpha > 0$ the Gaussian quantile and the position evaluated at the equilibrium aggregate σ_X , the constraint becomes $f \sigma_X \leq mW/z_\alpha$. Using $\sigma_X = \sqrt{S + DF^2}$ and imposing symmetry $f = F$,

$$F^2(S + DF^2) \leq \left(\frac{mW}{z_\alpha}\right)^2 \implies F \leq \bar{F}_{\text{VaR}}(g, \tau, m) \equiv \sqrt{\frac{-S + \sqrt{S^2 + 4D(mW/z_\alpha)^2}}{2D}}. \quad (34)$$

The constrained equilibrium is then $F_{\text{VaR}}^{\text{CE}}(g, \tau, m) = \min\{F^{\text{CE}}(g, \tau), \bar{F}_{\text{VaR}}(g, \tau, m)\}$. Since $\partial \bar{F}_{\text{VaR}} / \partial m > 0$, tighter margins (lower m) weakly reduce inflows even when the unconstrained equilibrium would be large. The instrument is most effective in the high-inflow regime, where the endogenous jump term DF^2 makes σ_X steep in F ; this is precisely the region in which Proposition 4 shows the competitive equilibrium over-invests, thus a fixed-rule margin constraint is automatically state-dependent and leans against the inefficiency.